

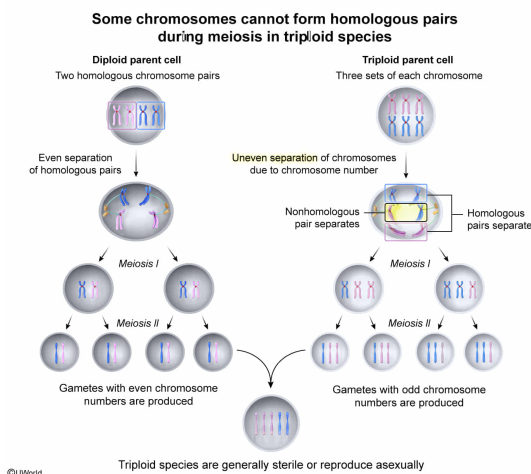
A certain eukaryotic species is triploid and has 24 total chromosomes. A feature of this species is:

- ✔ ☒ A. an inability of some chromosomes to form homologous pairs during meiosis. [64%]
- ☐ B. high fitness relative to diploid eukaryotic species. [12%]
- ☐ C. production of viable gametes with low genetic diversity. [11%]
- ☐ D. somatic cells with 12 chromosome pairs. [11%]

Correct

64% answered correctly

Explanation:



Ploidy refers to the total number of **homologous chromosome sets** in a cell nucleus. Triploid ($3n$) organisms have three complete sets of chromosomes instead of the two sets found in diploid ($2n$) organisms.

Triploid organisms may form via **nondisjunction** of *all* homologous chromosome pairs during **meiosis I** in a parent organism. This nondisjunction produces a diploid ($2n$) gamete (egg or sperm) that can be fertilized with another haploid ($1n$) gamete to produce a **triploid ($3n$) offspring** containing **three copies** of each chromosome.

Triploid organisms are generally sterile or reproduce **asexually** because they have an extra set of chromosomes that are unable to form homologous pairs during **meiosis**. As a result, triploid organisms are unable to produce gametes with even chromosome numbers.

In this example, a certain **eukaryotic** species is triploid and has 24 total chromosomes (ie, 8 groups containing 3 chromosomes each). As such, 16 of the 24 chromosomes can form homologous pairs during meiosis (ie, 8 pairs), but the remaining 8 chromosomes exist in the unpaired form. Therefore, a feature of this species is an **inability** of some chromosomes to **form homologous pairs during meiosis**.

(Choice B) Because triploid organisms are generally either sterile or reproduce asexually, any offspring they form are likely identical to the parent organism. Therefore, this species exhibits **low genetic diversity**, which decreases (*not* increases) its **fitness** (ie, reproductive success).

(Choice C) This species likely exhibits low genetic diversity due to its inability to **reproduce sexually**. Accordingly, the gametes it produces are *not* viable.

(Choice D) Because this species contains 24 total chromosomes and is triploid, its somatic cells have 8 groups of 3 chromosomes, *not* 12 groups of 2 chromosomes (ie, not 12 chromosome pairs).

Educational objective:

Ploidy refers to the total number of homologous chromosome sets in a cell nucleus. Triploid ($3n$) organisms have three complete sets of chromosomes, whereas diploid ($2n$) organisms have two complete sets of chromosomes.

Time spent:0 sec

Subject:
Biology

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System:
Reproduction